

定理 (任意一族环的直积的泛性质刻画) $(R_i)_{i \in I}$ 是任意一族环,
 R 是一个环, $(p_i: R \longrightarrow R_i)_{i \in I}$ 是一族映射, 满足:

* 每个 p_i 都是环同态

* 对 $\forall i \in I$, 任取 $r_i \in R_i$, 存在 $\alpha \in R$, 使得对 $\forall i \in I$, 都有 $p_i(\alpha) = r_i$

* 对 $\forall x, y \in R$, 若对 $\forall i \in I$, 有: $p_i(x) = p_i(y)$, 则有: $x = y$

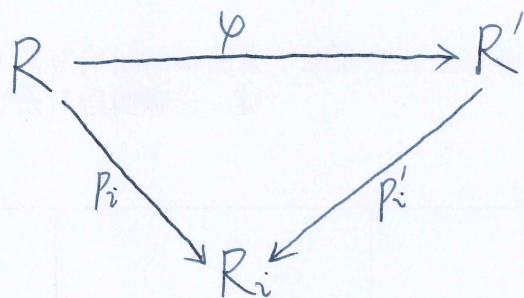
R' 是另一个环, $(p'_i: R' \longrightarrow R_i)_{i \in I}$ 是另一族映射, 满足:

* 每个 p'_i 都是环同态

* 对 $\forall i \in I$, 任取 $r_i \in R_i$, 存在 $\alpha' \in R'$, 使得对 $\forall i \in I$, 都有 $p'_i(\alpha') = r_i$

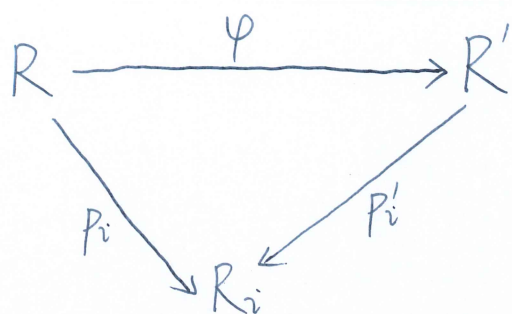
* 对 $\forall x, y \in R'$, 若对 $\forall i \in I$, 有: $p'_i(x) = p'_i(y)$, 则有 $x = y$

则有: 存在唯一的映射 $\varphi: R \longrightarrow R'$, 使得对 $\forall i \in I$, 下图交换:



且映射 φ 是环同构.

Proof: 首先, 忘记所有环的环结构, 仅仅把所有的环看作普通的集合,
可以得到: 存在唯一的映射 $\varphi: R \longrightarrow R'$, 使得对 $\forall i \in I$, 下图交换:



且映射 φ 是双射.

下证 $\varphi: R \longrightarrow R'$ 是环同态.

对 $\forall x, y \in R$, 有:

$\because x \in R, (p_i: R \longrightarrow R_i)_{i \in I}$ 是一族映射

$\because$ 对 $\forall i \in I, p_i(x) \in R_i \quad \therefore \exists \alpha' \in R', \text{ s.t. 对 } \forall i \in I, \text{ 都有 } p'_i(\alpha') = p_i(x)$

$\therefore \varphi(x) = \alpha' \in R'$

$\because y \in R, (p_i: R \longrightarrow R_i)_{i \in I}$ 是一族映射

$\because$ 对 $\forall i \in I, p_i(y) \in R_i \quad \therefore \exists \beta' \in R', \text{ s.t. 对 } \forall i \in I, \text{ 都有 } p'_i(\beta') = p_i(y)$

$\therefore \varphi(y) = \beta' \in R'$

$\because R$ 是环 $\therefore x+y \in R \quad \therefore \varphi(x+y) \in R' \quad \because xy \in R \quad \therefore \varphi(xy) \in R'$

$\because R'$ 是环 $\therefore \varphi(x) + \varphi(y) \in R', \varphi(x)\varphi(y) \in R'$

$\varphi(x) + \varphi(y) \in R', \varphi(x+y) \in R',$ 对 $\forall i \in I$, 有:

$p'_i(\varphi(x) + \varphi(y)) = p'_i(\alpha' + \beta') = p'_i(\alpha') + p'_i(\beta') = p_i(x) + p_i(y)$

$= p_i(x+y) = (p'_i \circ \varphi)(x+y) = p'_i(\varphi(x+y)) \quad \therefore \varphi(x) + \varphi(y) = \varphi(x+y)$

$$\therefore \varphi(x+y) = \varphi(x) + \varphi(y)$$

$$\varphi(xy) \in R', \varphi(x)\varphi(y) \in R', \text{ 对 } \forall i \in I, \text{ 有:}$$

$$\begin{aligned} p'_i(\varphi(xy)) &= (p'_i \circ \varphi)(xy) = p_i(xy) = p_i(x)p_i(y) = p'_i(\alpha')p'_i(\beta') \\ &= p'_i(\alpha'\beta') = p'_i(\varphi(x)\varphi(y)) \end{aligned}$$

$$\therefore \varphi(xy) = \varphi(x)\varphi(y)$$

$$\therefore 1_R \in R, (p_i: R \rightarrow R_i)_{i \in I} \text{ 是一族映射}$$

$$\therefore \text{对 } \forall i \in I, p_i(1_R) \in R_i \quad \therefore \exists \gamma' \in R', \text{ s.t. 对 } \forall i \in I, \text{ 都有 } p'_i(\gamma') = p_i(1_R)$$

$$\therefore \varphi(1_R) = \gamma' \in R'$$

$$\varphi(1_R) \in R', 1_{R'} \in R', \text{ 对 } \forall i \in I, \text{ 有:}$$

$$p'_i(\varphi(1_R)) = (p'_i \circ \varphi)(1_R) = p_i(1_R) = 1_{R_i} = p'_i(1_{R'})$$

$$\therefore \varphi(1_R) = 1_{R'}$$

$$\therefore \varphi: R \rightarrow R' \text{ 是环同态.}$$

$$\therefore \varphi: R \rightarrow R' \text{ 是环同构.}$$

